



ChronoLog MVP

Master's Thesis Project Defense (TFM)
Professional Appointment & Session Notes Management Platform

Hexagonal Architecture · Spec-Driven Development · Clean Code & SOLID

Author: Master's Student | Release v1.0.0
Official Repository: github.com/liandir-heriana/chronolog/tree/main

EXECUTIVE SUMMARY & METRICS

ChronoLog achieved 100% MVP completion with 24 verified tasks and 239 tests passed

24 / 24

Tasks Completed

100% of backlog verified
under SDD
methodology

239

Automated Tests

100% green with zero
failures, skips, or
regressions

86.40%

Global Coverage

95% coverage across
Core Domain & Use
Cases

v1.0.0

Production Release

E2E verified in Docker
Compose container
stack

CONTEXT & MOSCOW SCOPE

Clinical management required isolating data per practitioner and automated conflict-free scheduling

MUST (Mandatory)

- User Authentication & Authorization
- Client record registration & management
- Conflict-free appointment scheduling
- Persistent session & clinical notes
- Strict Anti-IDOR isolation (user_id)

SHOULD (Desirable - Included)

- Operational dashboard with daily metrics
- Human-readable client selectors (Dropdowns)
- Unified chronological history per patient
- Automatic session cleanup & sanitization

COULD / WON'T (Future Phases)

- Asynchronous Email / SMS notifications
- PDF export of patient clinical files
- Automated session summaries via LLM
- Postponed deliberately to protect core domain

ARCHITECTURAL DESIGN

Hexagonal Architecture decoupled pure Python domain logic from frameworks and database

1. Core Domain (domain/) — Pure Python · Zero Dependencies

- Pure entities: User, Client, Appointment, AuthSession
- Business invariants & schedule overlap rules
- Custom domain exceptions
- Fully independent of database or UI

2. Use Cases (use_cases/) — Application Layer · Orchestration

- RegisterUser, AuthenticateUser, RegisterClient
- ScheduleAppointment, CompleteAppointment
- Mandatory AuthenticatedUserContext injection
- Strict compliance with Dependency Inversion (DIP)

3. Infrastructure & UI — Primary & Secondary Adapters

- Postgres repositories with psycopg2-binary driver
- Interactive web interface built with Gradio UI
- Isolated containerization with Docker Compose
- Single composition root in main.py / composition.py

SECURITY BY DESIGN (DEFENSE-IN-DEPTH)

Mandatory user_id filtering and ORM-free queries eliminated IDOR and SQLi risks

[Security] Anti-IDOR Protection

Mandatory WHERE user_id = %s filtering across 100% of SQL persistence reads and mutations. Cross-user data access is impossible.

[Security] Robust Authentication

PBKDF2-HMAC-SHA256 password hashing, session tokens with strict expiration (expires_at), and Rate Limiting against brute-force attacks.

[Security] Parameterized SQL (No ORM)

Direct execution using psycopg2-binary with parameterized SQL queries. Eliminates SQL Injection vectors and avoids N+1 query bottlenecks.

[Security] Clean SAST Audit

Static security analysis with bandit, ruff, and mypy executed on every build. Zero High or Critical severity vulnerabilities.

SDD METHODOLOGY & WORKFLOW

Spec-Driven Development (SDD) structured the multi-agent orchestration

1

Phase 1: Spec & Proposal

Requirements, invariants, and architecture defined in proposal.md.

2

Phase 2: Task Breakdown

Granular breakdown of 24 tasks with specialized role assignments.

3

Phase 3: Multi-Agent Execution

Delegation to @backend-dev, @security-auditor, @frontend-dev, and @qa-tester.

4

Phase 4: TDD RED-GREEN

Writing failing tests prior to implementing any production code.

5

Phase 5: Quality Gate

Strict verification of coverage, ruff, mypy, and bandit before closing tasks.

TESTING STRATEGY & QUALITY GATES

The automated test suite achieved 86.40% global coverage and 95% across the core business domain

95.00%

CORE Layer (Domain & Use Cases) — Target 100% / Actual 95%

Unit tests covering entities, business invariants, and application use cases.

82.50%

IMPORTANT Layer (Presentation / Gradio) — Target 80% / Actual 82%

Tests covering UI handlers, navigation shell, and session management flows.

88.10%

INFRASTRUCTURE Layer (Adapters / Drivers) — Target 80% / Actual 88%

Integration tests against live PostgreSQL container instances.

86.40%

TOTAL APPLICATION (src/) — DoD $\geq$ 85% PASSED

239/239 tests passed with zero failures, skips, or regressions.

REDESIGN & USER EXPERIENCE (UX)

The Gradio interface integrated login-protected navigation and UUID-free client selection

[UX] Auth-Gated UI Shell

Authentication-protected interface. Main application tabs (Dashboard, Clients, Agenda, History) remain hidden (visible=False) until login.

[UX] Human-Readable Dropdowns (gr.Dropdown)

Total elimination of manual UUID copy-pasting. Forms display client names and surnames filtered for the active practitioner.

[UX] Operational Dashboard

Post-login landing screen displaying daily appointment metrics, active client counters, and quick action shortcuts.

[UX] Logout & State Sanitization

Immediate invalidation of session tokens, state variables, and data tables upon logout to prevent cross-practitioner leaks.

INFRASTRUCTURE & LIVE DEMO

Docker Compose orchestrates the stack and db/seed_demo.py enables instant live demonstrations

Containerized Deployment (Docker Compose)

- App container: Python 3.12 + Gradio UI exposed on port :7860
- DB container: PostgreSQL 16 with persistent volume
- Isolated Docker network for direct inter-container communication
- Active healthchecks configured on both services

Demonstration Data (db/seed_demo.py)

- Instant database seeding via python db/seed_demo.py
- Creates demo user account: demo@chronolog.com / Password123!
- Generates 3 test clients and historical appointments with clinical notes
- Guarantees a smooth demo without relying on manual data entry

CONCLUSIONS & FUTURE ROADMAP

Release v1.0.0 consolidates the production-ready MVP and sets the foundation for AI adapters

Key Achievements (Release v1.0.0)

- MVP 100% completed (24/24 tasks verified)
- Decoupled and robust Hexagonal Architecture
- Anti-IDOR isolation and security audit clean
- 239/239 tests passed (86.40% global coverage)
- Docker deployment with verified E2E test suite

Future Evolution (Roadmap)

- LLM adapter for automated post-consultation summaries
- Clinical history and patient file export to PDF
- Asynchronous Email & WhatsApp notifications
- Mobile adapter / Progressive Web App (PWA)
- Advanced scheduling & income analytics